

Hypothesis-Overview

Hypothesis testing compares two opposite ideas about a group of people or things using data from a small part of that group (A sample) to decide which idea is more likely true.

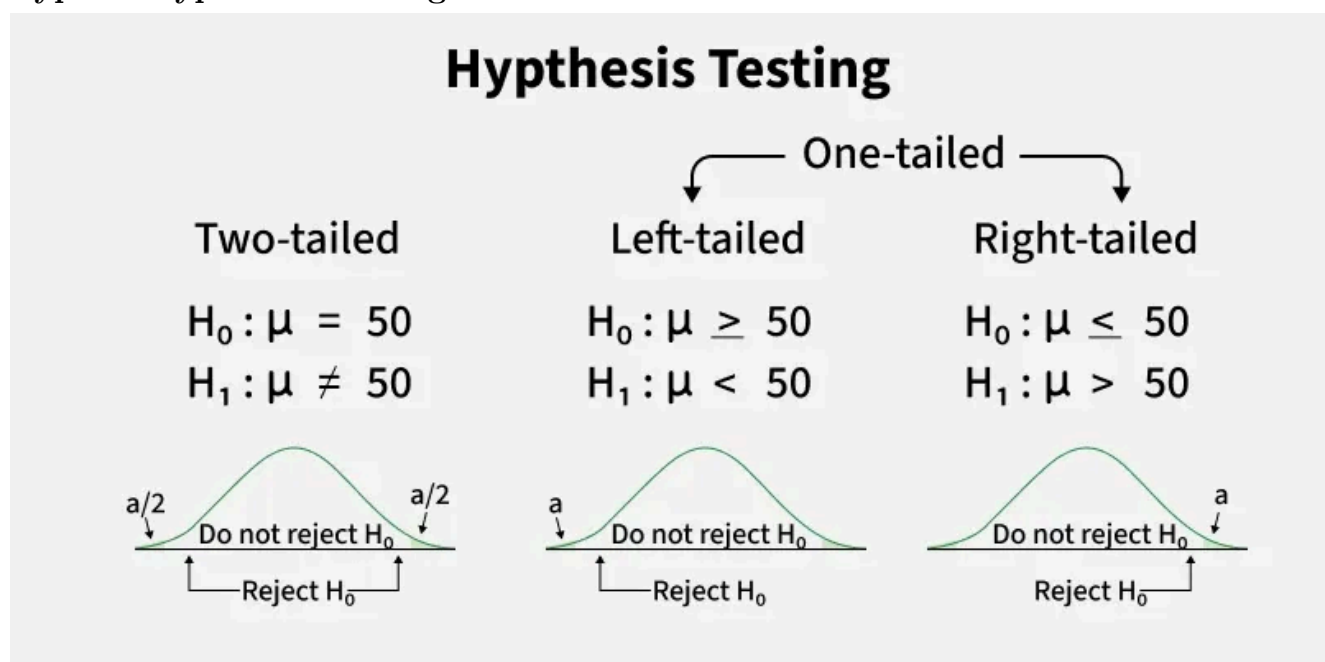
Defining Hypothesis:

- Null Hypothesis: the starting assumption (no effect or difference)
- Alternative Hypothesis: The opposite

Key terms:

- Significance level:** How sure we want to be before saying the claim is false. usually 0.05
- P-value:** The chance of seeing the data if the null hypothesis is true. if this is less than the significance level, we say the claim is false,
- Test statistic:** A number that helps us decide if the data supports or rejects the claim
- Critical value:** The cutoff point to compare with the test statistic
- Degree of freedom:** A number that depends on the data size and helps find the critical value

Type of Hypothesis testing



Type 1 and Type 2 Errors

	Null Hypothesis is True	Null Hypothesis is False
Null Hypothesis is True (Accept)	Correct Decision	Type II Error (False Negative)
Alternative Hypothesis is True (Reject)	Type I Error (False Positive)	Correct Decision

Make Decision

1. Using Critical Value:

- If test statistic $>$ critical value $\rightarrow$ reject H_0 .
- If test statistic $\leq$ critical value $\rightarrow$ fail to reject H_0 .

2. Using P-value:

- If p-value $\leq \alpha \rightarrow$ reject H_0 .
- If p-value $> \alpha \rightarrow$ fail to reject H_0 .

Difference Between One and Two-Tailed Test:

One-Tailed Test	Two-Tailed Test
A test of any statistical hypothesis, where the alternative hypothesis is one-tailed either right-tailed or left-tailed.	A test of a statistical hypothesis, where the alternative hypothesis is two-tailed .
For one-tailed, we use either $>$ or $<$ sign for the alternative hypothesis.	For two-tailed, we use $\neq$ sign for the alternative hypothesis.
When the alternative hypothesis specifies a direction then we use a one-tailed test.	If no direction is given then we will use a two-tailed test.
Critical region lies entirely on either the right side or left side of the sampling distribution.	Critical region is given by the portion of the area lying in both the tails of the probability curve of the test statistic.
Here, the Entire level of significance (α) i.e. 5% has either in the left tail or right tail.	It splits the level of significance (α) into half.
Rejection region is either from the left side or right side of the sampling distribution.	Rejection region is from both sides i.e. left and right of the sampling distribution.
It checks the relation between the variable in a singles direction.	It checks the relation between the variables in any direction.
It is used to check whether the one mean is different from another mean or not.	It is used to check whether the two mean different from one another or not.